

Dipayan Sarkar

✉ dipayanskr@gmail.com • 📞 +91 9735490241 • 🌐 dipayansarkar.com • 🐙 github.com/dipayanskr
👤 Google Scholar • 🆔 0000-0002-0765-5689 • 🗺️ huggingface.co/dipayanskr

Summary

Computational biologist and deep learning researcher building sequence-based models for protein interaction and design. Author of four papers on attention-based PPI prediction, uncertainty calibration, and generative protein binder design, and sole developer of three deployed model-backed web platforms serving trained models in production. Comfortable owning the full path from architecture and large-scale GPU/TPU training through to inference APIs and deployment.

Skills

Languages	Python, R, C, Bash, L ^A T _E X
ML / DL	PyTorch, Transformers & attention architectures, transfer learning, pre-training & finetuning, autoregressive generative models, uncertainty calibration, mixture-of-experts
Scale	Multi-GPU and HPC training, NVIDIA L4 / A100, Google Cloud TPU (v4, v5e, v6e)
Domain	Protein language models, protein-protein interaction & biomolecular networks, de novo binder design, biological sequence analysis
Data	NumPy, Pandas, scikit-learn, Matplotlib, Seaborn
Engineering	Docker, FastAPI, Flask, Nginx, Linux server deployment, Git, GitHub Actions, uv, Conda

Experience

PhD Research Scholar, Department of Bioinformatics, University of North Bengal Siliguri, India | 2022–Present

- Designed a transfer-learning hybrid attention architecture for sequence-based protein-protein interaction prediction, published in *BBA – Proteins and Proteomics*; released as a public web server.
- Built an uncertainty-aware calibrated ensemble for PPI network prediction in *Arabidopsis thaliana*, targeting reliable confidence estimates rather than raw accuracy alone.
- Developed a sparse mixture-of-experts autoregressive generative model for de novo protein binder design.
- Shipped and maintain three model-backed web platforms (Docker + FastAPI/Flask + Nginx) providing inference and interaction-network visualisation to external users.

Deployed Research Software

Deep-Interact Studio	Web platform to build, train, and compare deep learning models for protein-protein, drug-target, RNA-protein, and protein-DNA interaction prediction, with integrated interpretability. deepinteract.compbiosysnbu.in
AttnSeq-PPI	Deep learning protein-protein interaction prediction server. attnseqppi.compbiosysnbu.in
EGRNi	R package on CRAN (v0.1.6) for ensemble gene regulatory network inference. cran.r-project.org/package=EGRNi

Selected Publications

Sarkar, D., & Sarkar, C. (2025). AttnSeq-PPI: Enhancing protein-protein interaction network prediction using transfer learning-driven hybrid attention. *BBA – Proteins and Proteomics*, 1874(1), 141102. doi:10.1016/j.bbapap.2025.141102

Sarkar, D., & Sarkar, C. (2026). MoE-Bind: Guiding de novo protein binder generation with sparse experts. *bioRxiv*. doi:10.64898/2026.06.13.732043

Sarkar, D., & Sarkar, C. (2026). ARACoFusion: Uncertainty-aware calibrated deep learning for PPI network prediction in *Arabidopsis thaliana*. *bioRxiv*. doi:10.64898/2026.05.22.727120

Sarkar, D., Bardhan, K., & Sarkar, C. (2026). Deep-Interact Studio: An interactive deep learning model building platform for biomolecular interaction prediction. *bioRxiv*. doi:10.64898/2026.07.02.736034

Education

Ph.D. in Bioinformatics, University of North Bengal India | Expected submission 2027
Deep learning frameworks for predicting protein-protein interaction networks from protein sequences. Supervisor: Dr. Chiranjib Sarkar.

M.Sc. in Botany (Specialisation: Biochemistry), University of North Bengal India | 2021
77.25% – First Class.

Awards & Compute Grants

CSIR-UGC NET with JRF, Life Sciences (2021) – All India Rank 216. • **GATE** Life Sciences (XL), qualified (2022).

IndiaAI Compute Initiative (2025) – awarded NVIDIA L4 GPU cloud compute under the IndiaAI Mission for deep learning-based PPI network prediction (Sep 2025 – Aug 2026).

Google TPU Research Cloud (TRC) (2025) – awarded Cloud TPU access (v4, v5e, v6e) for machine learning research.